VERIDIAN PHARMA LTD. — Quality & Compliance Directorate, Hyderabad Facility

Standard Operating Procedure: Data Integrity and Governance

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Owner: Meera Rao (QA Head)	**Department:** Quality Assurance	**Status:** Effective

1. Purpose

This standard operating procedure mandates data governance requirements, electronic record controls, manual documentation standards, and audit trail review protocols across all cGMP systems at Veridian Pharma Ltd., Hyderabad.

Data integrity is the bedrock of pharmaceutical quality and regulatory compliance. Every decision regarding product release and patient safety relies upon authentic, contemporaneous, and uncorrupted data.

Follows ALCOA+ principles and satisfies FDA, MHRA, and WHO Data Integrity guidelines.

2. Scope

Applies across all computerized systems, analytical instruments, automated SCADA manufacturing networks, electronic spreadsheets, and paper records throughout the company.

Encompasses all employees, temporary contractors, data administrators, and external validation engineers.

Covers the full data lifecycle: generation, processing, review, reporting, archiving, and retrieval.

3. Responsibilities

• **All System Users:** Create and document raw data contemporaneously, maintain password secrecy, and report suspected data breaches or software glitches immediately.

• **IT Lead (Kiran Patel):** Implement role-based security access, maintain automated system time synchronization, and manage secure backup schedules.

• **QA Data Governance Lead:** Conduct scheduled audit trail reviews, perform data integrity risk assessments, and investigate potential audit trail falsifications.

4. Procedure

4.1 Data Governance Framework: Data governance controls must ensure data traceability, authenticity, and permanence across all operations.

4.2 Standard Compliance Principle: Records follow ALCOA+.

4.3 Periodic Surveillance Auditing: Audit trails reviewed monthly.

4.4 Strict Individual Accountability: Shared logins prohibited.

4.5 Unique Credential Provisioning: Every user must possess a unique username and cryptographically secure password to access computerized systems.

4.6 Anti-Deletion Software Protections: Electronic systems must prevent the unauthorized alteration, deletion, or overwriting of raw data files.

4.7 Automated Time Synchronization: Time synchronization must be automated across all servers and network-connected laboratory equipment to prevent manual backdating.

4.8 Paper Record Good Documentation Rules: Paper records must be contemporaneously signed and dated in indelible ink; correction fluid is strictly barred.

4.9 Escalation of Data Governance Breaches: Data integrity deviations or suspected falsifications must be reported directly to QA management for escalation.

Table 1: ALCOA+ Core Requirements and Technical Implementation

Principle	Operational Definition	Enforcement Control
Attributable	Traceable to individual performing activity	Unique user accounts, biometric or secure login
Legible / Original	Readable throughout lifecycle; first recording	Direct printout or secure electronic repository
Contemporaneous	Recorded at the exact time of execution	System automated timestamps, synchronised clock

5. Records and Documentation

Audit trail review certificates, user role authorization matrices, electronic backup logs, and annual data integrity self-assessment files must be archived.

6. Revision History

Version	Effective Date	Change Description
1.0	2026-07-30	Initial release of data integrity governance procedure.